

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

6cTDF4hLRDTcPodcf5uAAxDrYf51DSpJV92LiDorj2Cu

Date: 5/1/2025

Compounds: Amoxicillin, Bismuth Subsalicylate, Paracetamol, Aspirin

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Amoxaspiribis

Formula: C17H19N3O7S

Weight: 393.42

Drug-likeness: 0.7

Activity:

Broad-spectrum antibacterial and anti-inflammatory

Target Analysis

Penicillin-binding proteins

Binding Affinity: -7.2

High in treating bacterial infections

Cyclooxygenase-1 (COX-1)

Binding Affinity: -8.1

Moderate in reducing inflammation and pain

Toxicity Profile

Risk Level: Moderate

Affected Systems: Liver, Gastrointestinal Tract

Mitigation Suggestions:

- Dosage control
- Hepatic function monitoring

Pharmacokinetics

Absorption:

Moderate, oral with rapid onset

Distribution:

Wide, crosses the blood-brain barrier

Metabolism:

Hepatic

Excretion:

Renal

Half-life: 2-6 hours

Detailed Analysis

Amoxaspiribis is a novel compound hypothesized by combining the active functionalities of Amoxicillin, Aspirin, and Bismuth Subsalicylate, with influences from Paracetamol. The resulting structure aims to harness broad-spectrum antibacterial properties from Amoxicillin and anti-inflammatory effects from Aspirin. Bismuth's gastroprotective properties were considered to mitigate the gastrointestinal irritation typically associated with Aspirin, while the analgesic aspect of Paracetamol complements the compound's intended multifunctionality. Preclinical analysis suggests effective binding to penicillin-binding proteins and COX-1 enzyme, indicating potent antibacterial and anti-inflammatory activities. The compound's predicted pharmacokinetics support its use in acute settings, with rapid absorption and distribution. Moderate toxicity risks are associated mostly with metabolic and hepatic processes, with careful dose management recommended. Overall, the research supports further development, with planned investigations into its action mechanism, any unforeseen synergistic effects, and long-term safety.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>